Settings in a Tab: Click Heatmap Report

Click data analysis of the Lyssna prototype test · October 2026

Introduction

For this task I analysed the click data from an unmoderated Lyssna test of a new settings prototype. 76 developers had to open the settings, turn on “Use contrast scrollbars” and then go back to the code. The new design has two changes compared to the old one:

1. Settings open as a tab in the editor, not in a separate window.
2. When you change something, a popup asks you to save or revert, instead of the usual OK / Cancel / Apply buttons.

The short version of what I found: almost everyone managed to turn on the setting and save it, but a lot of people got stuck after that. 60 of 76 people (79%) saved the change, but Lyssna only counted 19 (25%) as successful.

Data and how I analysed it

The export had 76 respondents, one task each, and 851 clicks on six screens of the prototype:

Screen ID	Screen	What the user should do
1:12610	Main Window	Click the gear icon (top right)
1:12676	Context menu	Choose “Settings…”
1:12541	Default	Tick “Use contrast scrollbars”
1:12401	Changed	Click “Save Changes” in the popup
1:12472	Saved	Go back to the code (FunctionUtils.java tab)
1:12624	Code	Goal screen, no clicks here

I wrote a small Python project for this (NumPy, SciPy, Matplotlib and Pillow), split into a few files:

- study.py has a Study class that loads the JSON, collects the clicks per screen and calculates the stats (funnel, where people gave up, misclicks, first clicks, median times),
- heatmaps.py draws the heatmaps,
- saliency.py compares the first clicks with a simple saliency model (more about that below),
- screens.py has the screen ids and finds the right screenshot for each screen,

- main.py runs everything.

For each screen, the script counts the clicks on every pixel, smooths them with a Gaussian filter (sigma is 1.5% of the screen width) and puts the result on top of the screenshot. It makes two heatmaps per screen: one with all clicks and one with only the first click of each person. The click coordinates were already in the same pixels as the screenshots, so I didn't have to rescale anything. For every screen I also counted how many people hit the right spot with their first click.

There was also one more screen (1:12650) that only 2 people reached, with just one click in total, so I left its heatmap out of the report because it doesn't really show anything. From the paths it looks like it's the "unsaved changes" state you get if you try to leave settings without saving: both people got there from the "Changed" screen.

Overall results

Overall 69% of all clicks (587 of 851) were misclicks. Not many people are lost until they save the change. The big drop happens after saving, when they have to leave the settings:

Screen	People who reached it
Main Window	76 (100%)
Context menu	68 (89%)
Default (settings open)	68 (89%)
Changed (checkbox ticked)	64 (84%)
Saved	60 (79%)
Code	19 (25%)

The reason the Lyssna number is so low is partly how success was defined. It only counts if you get to the "Code" screen by clicking the code tab. 24 people saved and then closed the Settings tab instead, which is a pretty normal thing to do in an IDE, but the prototype sends you back to the first screen when you do that, so Lyssna counted them as "gave up". If I count them too, 40 of 76 people (53%) actually finished the whole task.

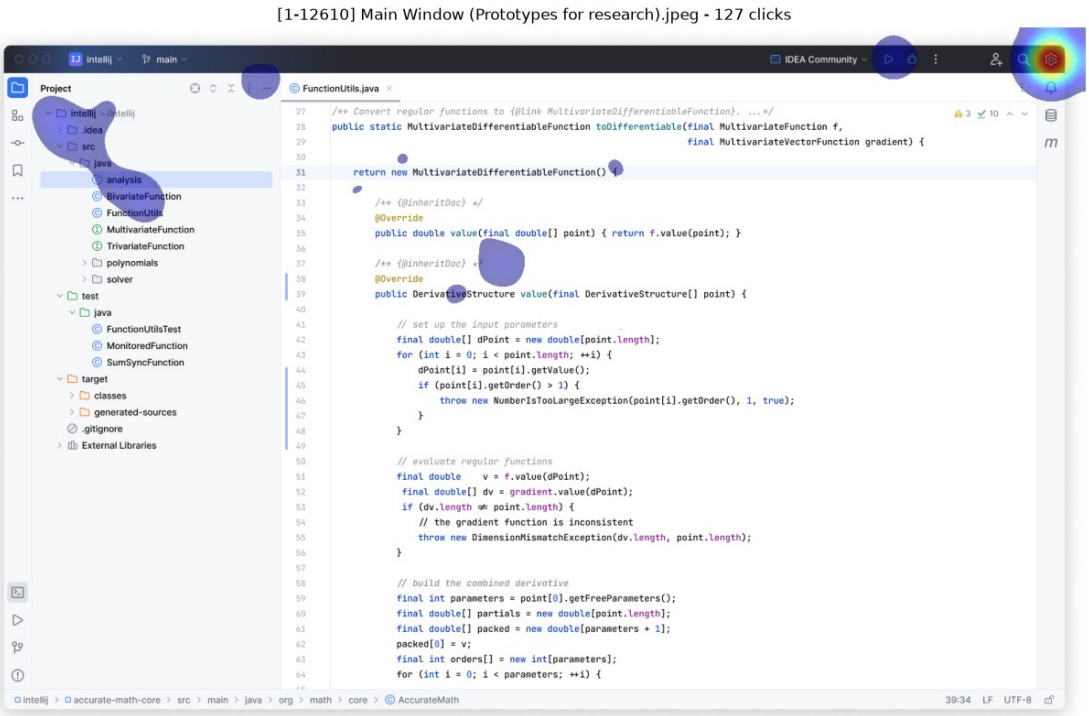
Of the 20 people who saved but didn't get back to the code, 17 just ended the task while still on the Settings tab, and the other 3 went back and forth a bit before giving up. Of the 16 people who never saved, 8 never even opened the gear menu and 5 opened the settings but didn't save. The other 3 are actually in the "completed" group, they switched to the code tab without saving. The median time was 77 s for people who completed and 58 s for the ones who gave up.

Findings per screen

In all the heatmaps, the warmer the color, the more clicks there were in that spot. The number of clicks is in the title of each image. Each heatmap is scaled to its own maximum, so the colors can't be compared between screens.

1. Main window

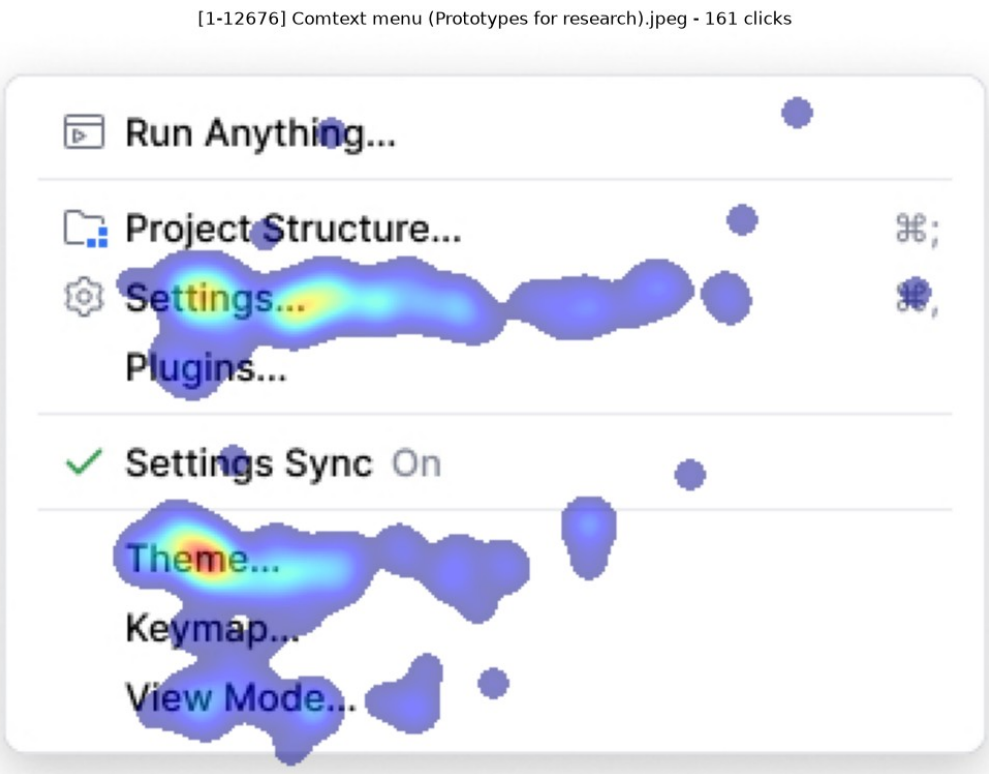
The gear is the hottest spot, but only 32 of the 59 people with recorded clicks clicked it first, and 8 of 76 never opened the menu at all. The other clicks are mostly in the code editor and the project tree, and a few people also clicked the Run and Debug icons next to the gear.



Main window heatmap

2. Gear menu

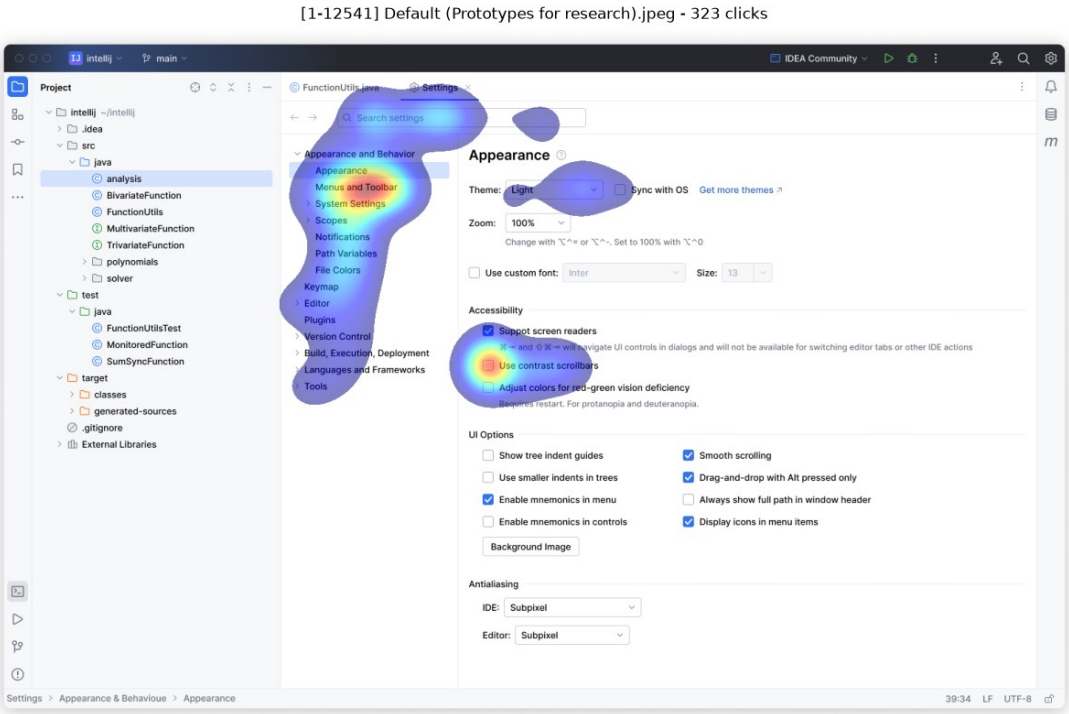
43 of 68 people (63%) clicked “Settings…” first. The other big hot spot is “Theme…”, and there are also some clicks on “View Mode…”. I think this is because the task said “change the appearance”, and “Theme” just sounds more like appearance than “Settings” does.



Gear menu heatmap

3. Settings page

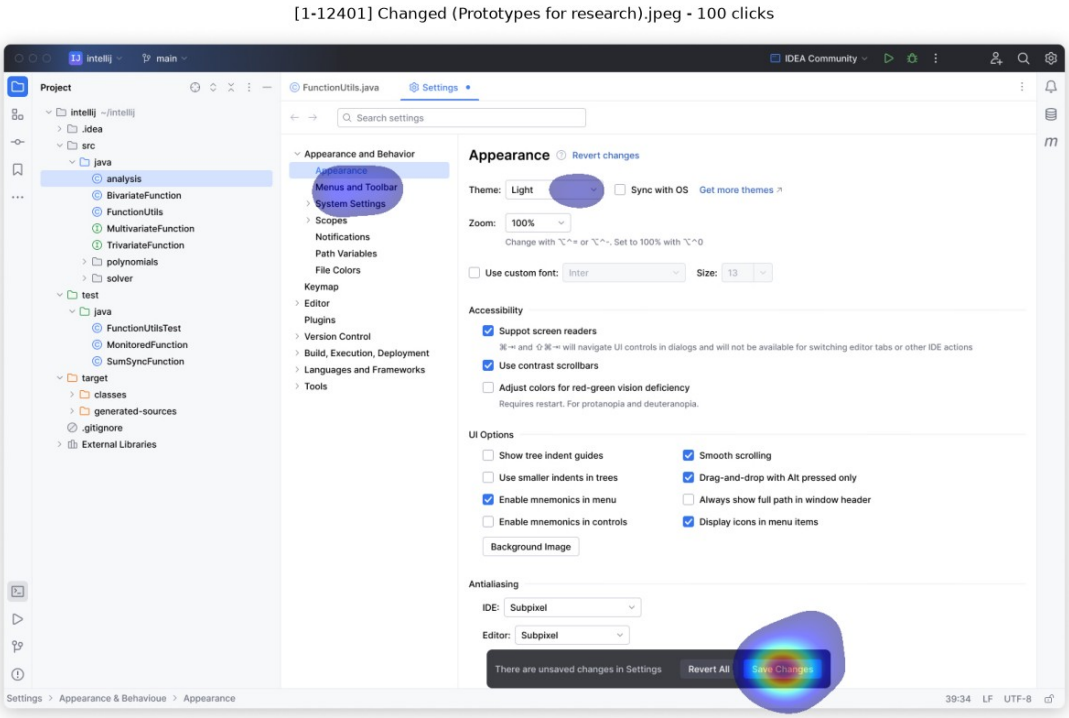
This screen had the most clicks (323). The settings already open on the Appearance page and the checkbox is visible right away, but only 21 of 65 people clicked it first. Most of the other clicks went to the category tree on the left and to the search bar. Honestly that's what I would do too in a real IDE, but in the prototype neither of those works, so it all counts as misclicks.



Settings page heatmap

4. After ticking the checkbox

This part went well. 57 of 64 people (89%) clicked “Save Changes” first, so the popup is easy to notice. On the heatmap there are almost no clicks around the popup except on the Save button, and only a few on the settings tree and the theme dropdown.

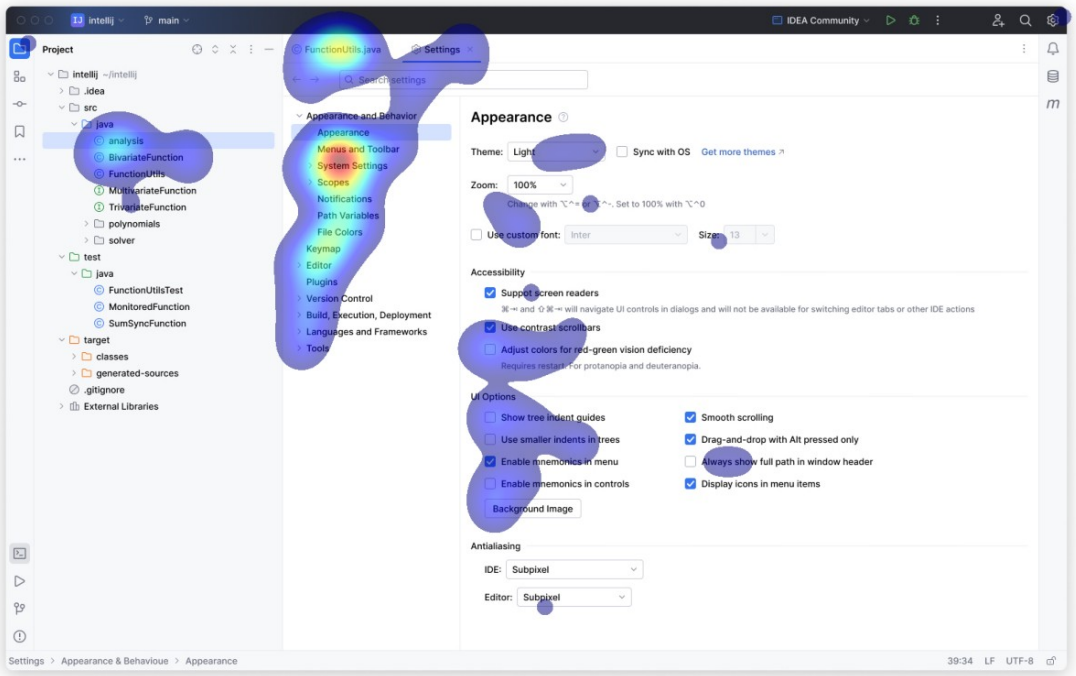


Changed state heatmap

5. After saving

This is where most people got lost. The way back to the code is the FunctionUtils.java tab, and it is the hottest spot, but only 12 of the 25 people who clicked on this screen hit a hotspot first. The others clicked around in the category tree, the project tree and the settings. Most people who got here didn't click anything at all, they either closed the Settings tab or just ended the task.

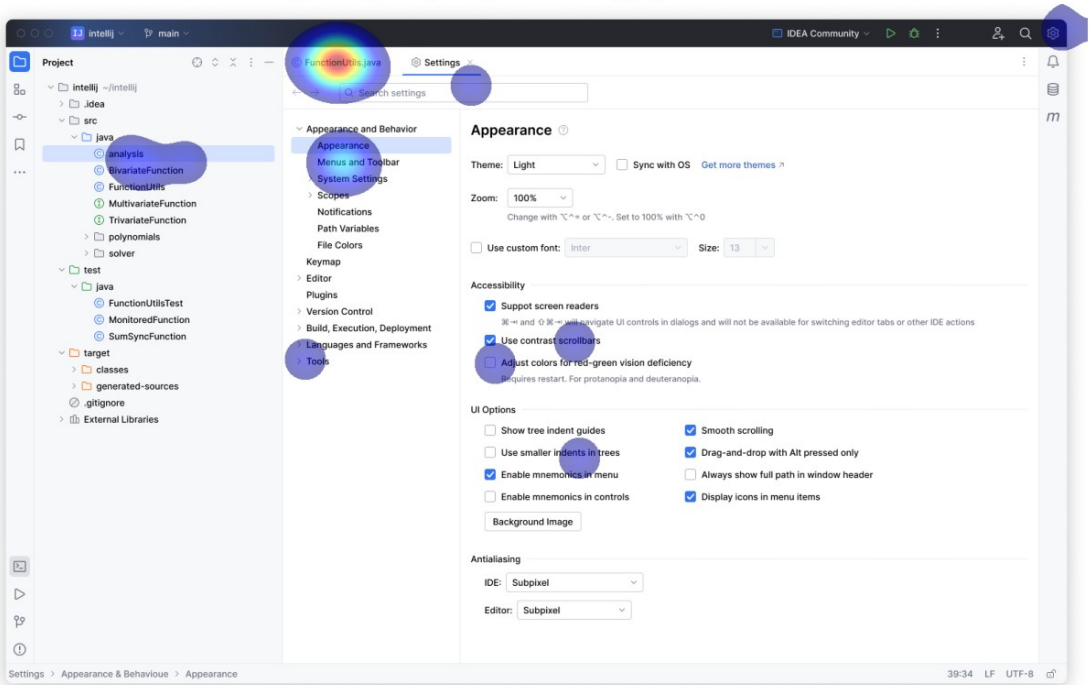
[1-12472] Saved (Prototypes for research).jpeg - 139 clicks



Saved state heatmap

The first-click heatmap shows it more clearly: of the people who clicked here, most first clicks went to the FunctionUtils.java tab, so the ones who looked for the way back mostly found it.

[1-12472] Saved (Prototypes for research).jpeg - 25 first clicks



Saved state, first clicks only

Do first clicks follow what stands out on the screen?

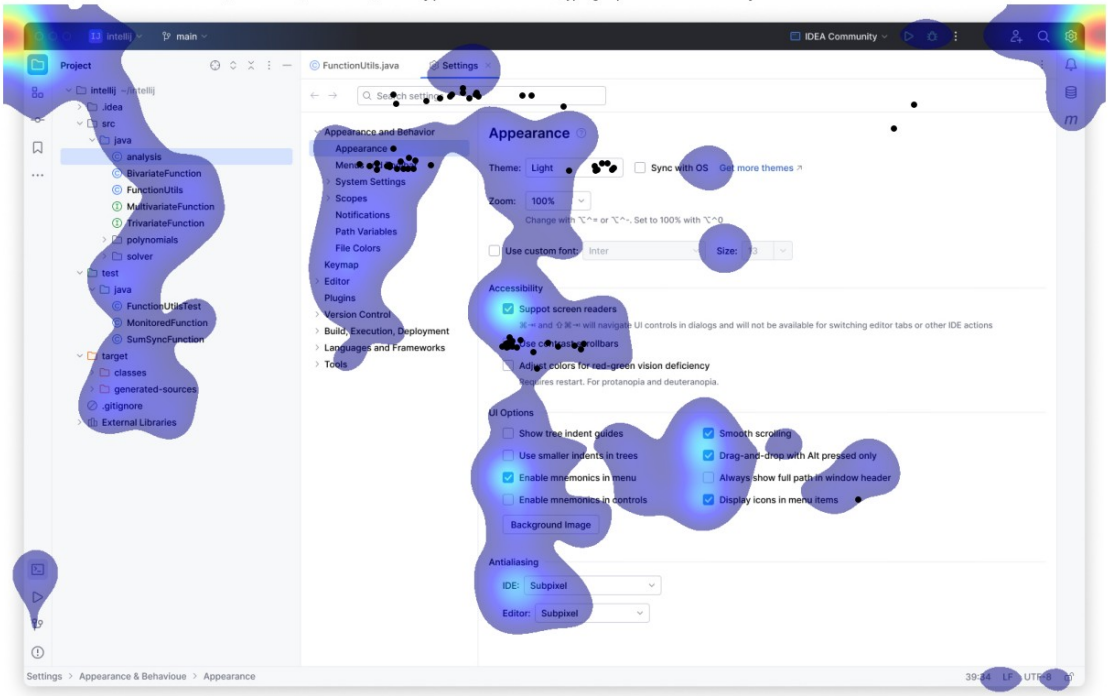
I also wanted to see if people click first on the parts of the screen that stand out visually, or if they go to what they are looking for. For this I made a saliency map of every screenshot with the spectral residual method (Hou & Zhang, 2007). It's a simple model that marks the parts of an image that "pop out". Then I checked how many first clicks land in the top 20% of the map. As a comparison I did the same with a plain edge map (where there is a lot of contrast), to see if the saliency model adds anything. If clicks were completely random, about 20% would land there.

Screen	First clicks	In top 20% (saliency)	In top 20% (edges only)
Main Window	59	64%	69%
Context menu	68	43%	43%
Default	65	72%	54%
Changed	64	92%	94%
Saved	25	44%	92%

On most screens first clicks land in the "visible" parts much more often than by chance, but saliency isn't really better than just edges. The only screen where it helps is Default, where the checkbox area stands out (72% vs 54%). On the Saved screen it's the opposite: the first clicks went to the tabs at the top, which have a lot of edges but the saliency model doesn't mark them. In the menu, both maps mark all the menu items the same, so they can't say why people picked "Theme…" over "Settings…". I left out the 1:12650 screen because it has only one first click.

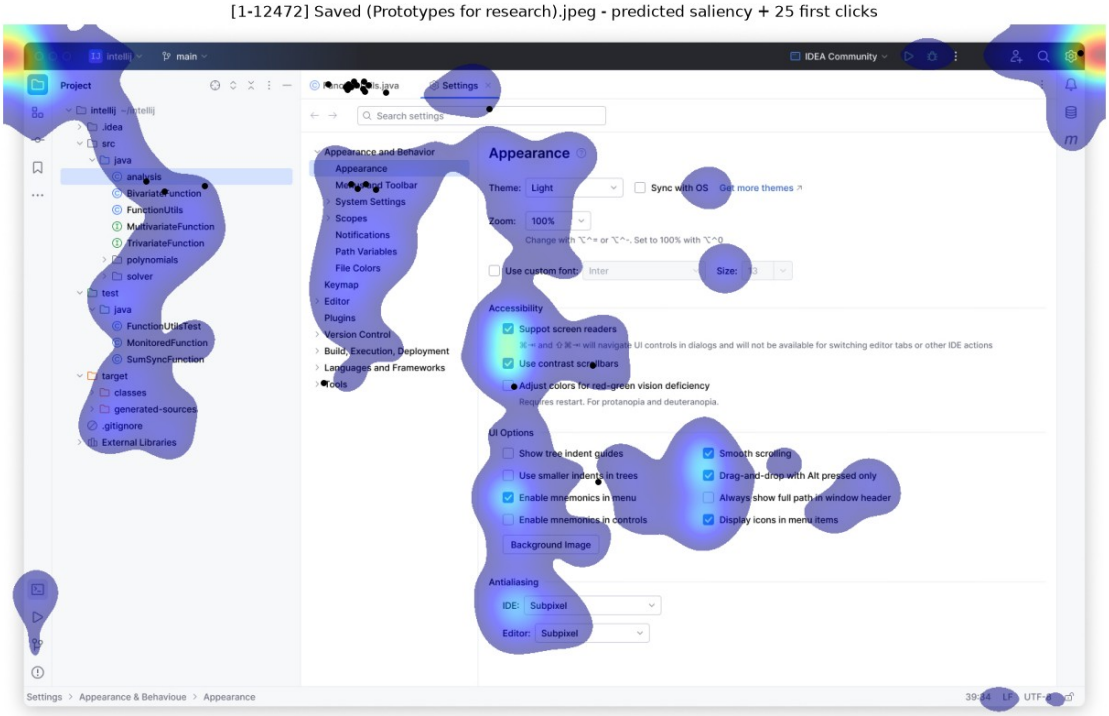
So my conclusion is that people don't just click on what stands out, they mostly go to what fits the task. Visual saliency can't explain mistakes like "Theme…" or looking through the category tree.

[1-12541] Default (Prototypes for research).jpeg - predicted saliency + 65 first clicks



Saliency and first clicks on the Default screen

[1-12472] Saved (Prototypes for research).jpeg - predicted saliency + 25 first clicks



Saliency and first clicks on the Saved screen

What this means for the two changes

The save / revert popup (change 2) seems to work. 89% clicked “Save Changes” first, and there are almost no misclicks around the popup. So people notice it and understand it even without the OK / Cancel / Apply buttons.

Settings in a tab (change 1) is where the problem is, because there is no clear way out. In the old window OK saved and closed everything with one click. Now after saving you just stay on the Settings tab and closing it is a separate step that nothing tells you to do. I saw two main reactions:

- 17 people stopped right after saving, probably because they thought the task was done.
- 24 people closed the tab and went back to the code, which is the normal thing to do in an editor.

Recommendations

1. After “Save Changes” there should be a clear next step, for example a “Saved” message with a “Close settings” button, or the tab could close by itself if it was opened just to change one setting.
2. Show that the setting is now on (for example highlight the scrollbar for a moment), so people know the change worked and the task is done.
3. Make “Settings…” easier to tell apart from “Theme…” and “View Mode…” in the menu, or let “Theme…” also open the Appearance page.
4. For the next test, success should be “setting saved and back in the code”, and both ways of getting back (clicking the code tab or closing the Settings tab) should count. It would also help to make the category tree and search clickable, since that’s how developers usually look for settings.

Limitations

- It’s a prototype, not the real product. Only the hotspots do something, so search, the category tree and other controls don’t work. A lot of the misclicks on the settings screens are probably just normal behavior.
- For some people a screen change has no recorded click (for example the menu opened but there is no click on the gear), so the first-click numbers only use the clicks that were recorded.
- Clicks are saved per screen and not per visit. If someone was on the same screen twice, the clicks from both visits are mixed together, so I can’t always be sure how exactly they got back to the main window.
- The success definition is quite narrow. 3 of the 19 “completed” people never actually saved the setting.
- The saliency model is very simple and was made for photos, not for UI screenshots, so those numbers are just a rough check.

- There was no think-aloud and no follow-up questions, so the reasons why people stopped after saving are just my guesses from the paths. It would be good to check this in a few moderated sessions.